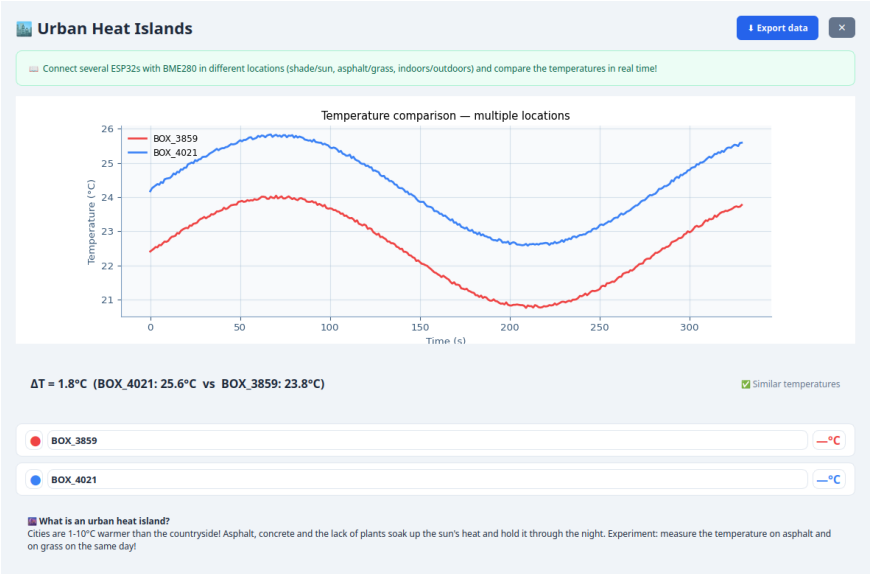
Heat Islands with Several Boxes

Several boxes at the same moment, and the limit below which a difference means nothing

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** 20 min, 50 min or a project of 4 h • **Sensors:** BME280 (temperature, humidity, pressure) on two or more boxes



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). The same app serves both levels, but the task differs: at **Explorer**, students compare ΔT with the co-location scatter and answer "detectable / not detectable"; at **Scientist**, they also subtract each box's individual offset and repeat the route at three times of day. The most direct curriculum entry points are **year V** (geography: the Earth's atmosphere), **year VIII** (physics: heat transfer) and **year IX** (general physical geography).

Assumed knowledge

The guided exercise *Urban Heat Islands* (single-box), done first: the field protocol comes from there. In addition: the difference of two decimals, the maximum and minimum of a set of numbers. For the project: the arithmetic mean and an XY chart in a spreadsheet.

Time in class

Short: 20 min. Full: 50 min. Project: three 50-minute lessons plus three 30-minute outdoor sessions, that is 240 minutes (4 h), spread over two to three weeks.

Teacher preparation

30 min — **the co-location test of every box**, ten minutes on the same table, out of the sun; labelling each box with the exact name it shows on screen; walking the route with a watch in hand; checking WiFi coverage at each location; printing the schoolyard plan.

Sensor settling time

Five minutes at each location, and **ten minutes** at the first move outdoors. A box brought from a 22 °C classroom into 8 °C air drifts for minutes on end. This is the single most common reason for poor data.

Sensors and boxes BME280 on **at least two boxes**. With only one, the ΔT card shows no number at all, only the message **Connect at least two boxes**. Four to six boxes make the survey much richer, because they let every location be measured *simultaneously*.

Three routes

Short 20 min: a projected demonstration — two boxes side by side on the same bench, then one on a sunlit sill and one in shade, and a discussion of what the gap between the two lines means.

Full 50 min: a single walk — briefing, co-location, four locations, export, chart and discussion. Needs a dry day and outdoor supervision.

Project Three 50-minute lessons plus three 30-minute outdoor sessions: the same route at three times of day, the location × time matrix, the heat map and the written recommendation.

The single-box exercise first, then the multi-box app

The two heat-island resources are a pair, and their order is not arbitrary. The guided exercise *Urban Heat Islands* (single-box) is done first. It is not a smaller version of this one, but the protocol lesson this app needs in order to mean anything: the same 1.2 m height at every location, a white screen above the sensor at every location including the shaded ones, the waiting time respected, a two-metre exclusion circle, the time noted with the number — and, above all, the idea that every measurement has a scatter of its own, worked out there from the difference between the two readings taken at the starting location.

What the app adds is simultaneity: several boxes read *at the same moment*, so the weather has no time to change between the second location and the fifth. And, with it, it adds what a single box cannot do: **measuring the detection limit directly**, by putting the boxes side by side before setting off.

A class handed six boxes before holding one still for five minutes at a measured height produces six data sets that cannot be compared, and no student will know which of the numbers means anything. If you do not have time for both, do the exercise: it has a complete conclusion, with no app needed. The app without the exercise does not.

1. What the app shows, and what it does not

Unlike the guided exercises, this is not a step-by-step wizard. It is an **open instrument**, with a single purpose: comparison. It plots the temperature of every connected box on the same axes and continuously reports the difference between the warmest and the coolest, which it judges against the **scatter** the class measured with the boxes side by side. It has no green button that advances, no **Back**, no **Finish** and no observation journal. **You set the pace of the lesson, not the app**, and the investigations in section 6 are ready to run exactly as written.

- **A single comparison chart**, with one coloured line per box, labelled with the box's name — the one given in **Settings**, or its identifier if you gave it none. Each box keeps its colour throughout the app.
- **The ΔT card**, below the chart, which shows the difference between the warmest and coolest box, names both of them, and adds a verdict in words that compares it with the scatter.
- **The device list**, on the right, with a coloured dot and the temperature of each box — the average of the last minute — so students can match a line on the chart to a box sitting at a particular location.
- **The Record the scatter button**, below the list, pressed while all the boxes stand side by

side; it keeps the difference between them as the detection limit.

- **A theory panel**, **What is an urban heat island?**, with the formula $\Delta T = T_{\max} - T_{\min}$ and the meaning of each symbol.
- **The Export data button**, which writes one dated file per box seen in the session, into a folder of your choice.

Technical note — what is really on screen, and what buttons actually exist

How to start it. From the **Activities** page, with the **Start** button on the **Heat Islands with Several Boxes** card — not the guided exercise **Urban Heat Islands**, which follows a single box. Because the app works with several boxes at once, **no device picker appears** and **it opens even if no box is connected**: the chart then reads **No box connected: switch on the boxes with a temperature sensor**, and the list, **No box connected yet**. It automatically takes *every* connected box that reports temperature. At the top, a green banner sums up the protocol: the boxes side by side for ten minutes, then **Record the scatter**, then different places, always in the shade, at the same height, never in the hand, five minutes at each place.
The chart. A single set of axes, **Air temperature**, in °C, with time on the horizontal axis in minutes and seconds (m:ss). One coloured line per box, with its name in the legend; each box has the same colour on every screen of the app. The chart refreshes every second and keeps the last 3600 readings of each box — about an hour. Above it sit the **Smooth**, **Pause** and **Clear** buttons (the last one starts the chart again from 0:00). A box that has disconnected stays on the chart, its line faded and thinner, and the legend adds **(offline)** to its name.
The device list. Each box gets a row with its coloured dot, its name and, on the right, **the average of the last minute** of its temperature.
The ΔT card. While at most one box is connected, it reads ΔT **= — °C | Connect at least two boxes**. From two boxes upward it shows the difference and the two extremes, with the boxes' names — for example ΔT **= 2.8 °C (Asphalt: 27.9 °C vs Grass: 25.1 °C)**. **All three numbers are averages over the last minute**, not instantaneous readings (a small note under the card says so: *Each value is the average of the last minute*), so a one-second spike no longer moves it — but after you move a box, the card needs at least a minute to show the new place. Under the number sits the verdict in words, described in the technical note at investigation 1.
The Record the scatter button. It is pressed once, at the end of the co-location, with every box side by side. The app keeps the difference between the warmest and the coolest box at that moment (again on the one-minute averages) and writes under the button, for example, *Scatter side by side: 0.3 °C. Differences smaller than this cannot be told apart.* Until the first press, the line reads *not measured yet.* With only one box connected, the button asks you to *Connect at least two boxes and stand them side by side.* The value is not kept between sessions: it is measured again every time the window opens — which is the rule anyway.
The picture and the data. The window has no **PNG** button: for the *picture* of the chart, take a screenshot; for the *data*, use the export.
The export. Export data, top right, next to the **Close** button. It first asks for a folder — the window is titled **Choose the export folder** — and then writes **one separate `.xlsx` file per box seen in the session**, named after the box's name and the date, for example `Asphalt_2026-09-14.xlsx`. The file holds the raw readings, not the smoothed chart line, with an `elapsed_s` column. On success, **Export complete** appears with the number of files written; if no box has been connected, **No box is connected.** appears instead. Put each outing's export in a folder named with its date and time, or a second walk on the same day will overwrite the files from the first.

How long to wait. Five minutes at each location, and **ten minutes** at the first move outdoors. The reading is taken at the end of the interval, not on arrival.

2. Where this fits in the curriculum

- **Geography** (*Geografie*, years V and IX) — at **year V**, the Earth's atmosphere and air temperature; at **year IX**, general physical geography covers the atmosphere, air temperature and climate, and the activity produces, with classroom instruments, exactly the quantities taught there. The fieldwork technique — sampling, route, representation on a map — is a geography skill at every level. The link to human settlements and urban planning, from **year VIII** and in upper secondary, is an *application* of the measured result, not a curriculum entry about the urban microclimate.
- **Physics** (*Fizică*, year VIII and upper secondary) — at **year VIII**, heat transfer by conduction, convection and radiation, plus specific heat: exactly the vocabulary the explanation needs. At **year VI**, temperature and thermal equilibrium. In upper secondary, the energy balance of a surface can be written out in full, with albedo and fluxes.
- **Biology** (*Biologie*) — plant transpiration and the role of vegetation. Cooling through evapotranspiration is a direct application, measurable with the boxes used here, but it remains an application, not a separate curriculum entry.
- **Mathematics** (*Matematică*, years V–VII and upper secondary) — the maximum, minimum and range of a data set at **years V–VI**; the arithmetic mean and plotting points on a coordinate system at **year VII**; several series on the same axes and comparing a difference with an uncertainty, in upper secondary. **Comparing an effect with the instrument's scatter is the cleanest meeting of school statistics with a real phenomenon in the whole collection.**
- **Computer science and ICT** (*Informatică și TIC*) — processing data in a spreadsheet: the handful of exported files line up by time of day, the mean of each series is computed, and an XY chart with several series is drawn. The exported files open directly.
- **Cross-curricular** — the distinction between *not detectable* and *does not exist*, the honesty of reporting a day when nothing was found, and the habit of measuring the instrument before measuring the world.

The transferable skill is **controlled comparison in the field**: two measurements that differ in a single thing, in a world that refuses to stand still. And the central intellectual move, one that will be reused in every physics laboratory and every biology report, is this: **before comparing two locations, measure how much the instruments themselves differ when placed at the same spot.** That number, not your wish, decides what you are allowed to claim.

3. Time plan

Route	Time	What happens
Short	**20 min**	A projected demonstration. Two boxes side by side on the same bench, five minutes, and reading the scatter off the ΔT card, with the **Record the scatter** button (7) — **this step is essential, and it is exactly the one that gets skipped**. Then one box on a sunlit sill and the other in shade, until the lines separate (8). Discussion: what produced the gap between the lines, and how many times bigger it is than the scatter (5). No fieldwork, no sheets.
Full	**50 min**	The safety briefing and the written route (7), co-locating every box on the same table (5), the walk — **four locations, five minutes each, that is** $4 \times 5 = 20$ **minutes** —, the return and the export (3), the chart and discussion, with ΔT compared with the scatter (15). Needs a dry day and outdoor supervision.
Project, lesson 1	**50 min**	The energy-balance theory, in words (12), the written hypothesis with a number, and discussing the reasons (10), the co-location test of every box, ten minutes on the same table, with the scatter recorded on the sheet (13), choosing the four to six locations on the schoolyard plan and the written route (15).
Project, three outings	3×30 min	The same route, at three times of day — early morning, at noon and at the end of the school day — ideally on the same day. Each outing: co-location (5), four locations of five minutes each (20), return and export (5).
Project, lesson 2	50 min	The location × time matrix (12), the three walks on the same axes (13), each walk's ΔT compared with the scatter (10), the warmest and coolest location (15).
Project, lesson 3	50 min	The schoolyard's heat map (15), the written recommendation, backed by a number from your own data (15), a critique of the method — what professional control was missing and how much difference it would have made — and the link to the town's heat island, measured at night between town and country (20).

The project means **three 50-minute lessons** and **three 30-minute outdoor sessions** — that is $3 \times 50 + 3 \times 30 = 240$ minutes, **four hours**, spread over two to three school weeks. The 50-minute full route is a single walk carried through to the end, and it is the right choice if you have only one window of time.

Give the boxes names you recognise, before going out

The chart legend and the ΔT card use the **box's name**, not the location where you placed it. Without a name, the box's identifier appears, an opaque string, and students will not know which line is the car park and which is the lawn. Name the boxes in **Settings**, under **Box names** — "Group 1", "Asphalt", "Grass" —, and the name is kept. And in addition, **write the name on a label stuck to the box** and on the clipboard of the group carrying it.

The same label solves the export problem too: the files come out named after the box, not the location, and a week later nobody remembers which box stood on the asphalt.

4. Materials

What you need	Per group	Notes
MagicBox with BME280	1	**At least two boxes in total**, or the ΔT card shows no number. Four to six make the survey much richer, because every location is measured at the same moment.
Laptop with the app	1 per class	For a walk on foot, a single laptop, carried by an adult, with all boxes on WiFi; for a comparison with fixed posts, the laptop stays in place.
Labels for the boxes	1 set	Each box's identifier, written large, stuck on the box and recorded on the group's sheet. Without them, the chart has lines with no names.
White screen for each box	1 / box	White card or a folded sheet. **All** boxes are shaded, or none are — **never mixed**, since then the measured difference is just a difference in treatment.
Small tripod, stick or crate	1 / box	So the sensor sits at the agreed height, not on the ground and not in a warm hand.
Tape measure	1	For the height above the ground. Matters much more than students expect.
Schoolyard plan, printed	1	A4 or A3, with the locations numbered. A satellite image makes the surfaces visible.
Clipboard and **pencil**	1	Pens fail in damp weather; pencils do not.
Watch or phone	1	The time of every reading. The sun's angle changes during the walk.
Route written in advance	1	The locations in a fixed order, with a time budget for each. Improvised routes produce unusable data.
Paired locations, in sun and in shade	4–8	Chosen deliberately in pairs: grass in sun and grass in shade, asphalt in sun and asphalt in shade. A pair answers a question; a single location answers none.
Reflective vests	1 / student	Compulsory if any location is near a car park, a service road or the school gate.
Ordinary thermometer, for cross-checking	1 (optional)	An independent instrument keeps everyone honest.

Preparation. Walk your own route first and time it: a route that looks like ten minutes on the map usually takes twenty-five. Check that WiFi reaches every location: the boxes record nothing on their own, so a place with no signal is a place with no data; if it does not reach, take the laptop closer or choose another place. **Most important: do the co-location test** — all boxes side by side, on the same table, out of direct sun, for ten minutes — and record how much they differ. A scatter of a few tenths of a degree is normal and must be subtracted from the field results; a scatter of over about a degree means a box must be dropped from the set or its offset noted and used as a correction. Finally, get approval for working outdoors, notify a second adult, and check the forecast: rain on the sensors ends the walk.

5. Scientific background for the teacher

The energy balance of a surface, and the urban heat island

Any surface is subject to an energy budget. The absorbed net radiation has to leave somehow, and the way it leaves decides how hot the surface becomes — and the air above it:

$$Q^* = Q_H + Q_E + Q_G,$$

where Q^* is the net radiation, Q_H the sensible-heat flux that warms the air, Q_E the latent-heat flux carried away by evaporation, and Q_G the heat conducted into the ground or the material. The net radiation itself is

$$Q^* = (1 - \alpha)\, S{\downarrow} \;+\; L{\downarrow} \;-\; L{\uparrow},$$

with $S{\downarrow}$ the received solar radiation, α the **albedo** (the reflected fraction) and L the infrared exchange.

A city changes almost every term. Dark asphalt has an albedo of about 0.05–0.15, so it absorbs 85–95% of the light that falls on it; fresh grass sits at about 0.20–0.25, and light concrete at about 0.30. Without plants and without damp soil, Q_E collapses towards zero, so almost all the absorbed energy has to leave through Q_H and Q_G — heating the air and charging the material. That stored heat is released after sunset, which is why the urban heat island is usually *strongest at night*, not at noon. Narrow streets trap infrared that would otherwise escape to the sky, and heat given off by vehicles, buildings and air-conditioning systems adds a further term. Typical differences between a city and the surrounding countryside are 1–3 °C by day and 3–10 °C on a clear, calm night.

5.1 Thermal mass: why the time of day matters more than students think

Concrete and asphalt have a specific heat around 800–900 $\mathrm{J\,kg^{-1}K^{-1}}$ and high densities, so a paved surface stores a lot of energy per square metre. It heats slowly in the morning, keeps warming all afternoon, and stays warm long after sunset. Grass on damp soil behaves differently, because a good part of the received energy leaves by evaporating water.

The practical consequence is large: **the hour at which you measure decides what you find**. At 9 o'clock the difference between grass and asphalt can be almost nil; the same route at 15 can give several degrees. **Repeating the route at different times is not an optional extension — it is the experiment itself.**

5.2 Evapotranspiration: the cooling plants provide

Vaporising one kilogram of water costs about 2.45×10^6 J at ordinary temperatures. A tree that transpires several hundred litres on a warm day removes from the system an amount of energy comparable to a continuously running air conditioner, and does so without electricity and without noise. Because this energy leaves through the Q_E term, it does not show up as a rise in temperature. This is why grass in full sun can be noticeably cooler than asphalt in full sun, even though they receive the same radiation — and why the difference shrinks in a drought, when the grass turns yellow and stops transpiring.

5.3 Air, not surface

The BME280 measures the temperature of the **air around it**, not of the surface it sits on. Sunlit asphalt can exceed 50 °C at the surface, while the air at 1.2 m reads 28 °C. Air is heated from below, by conduction and convection, so the difference between locations is always much smaller in air than at the surface, and it fades quickly with height.

This is exactly the conceptual trap of the guided exercise, and if the class has done it, they already know it. If not, say it *before* the first reading: otherwise students will expect the dramatic figures they have read about asphalt and will mistakenly mark a correct result as wrong. The consequence for this app is direct: **all boxes at the same height, measured with the tape**. See also investigation 5, which turns exactly this point into the thing being measured.

5.4 Co-location: where the detection limit comes from

Two boxes placed side by side, in the same air, will not show the same number. The difference between them is not a fault: it is the limited accuracy of each sensor, plus small differences in casing and exposure. **That is your detection limit, and it must be measured, not read off a data sheet**, because it depends on the units you own, their age and the temperature you work at.

How to measure it. All boxes side by side, on the same table, out of direct sun and draughts, five minutes for a routine outing and ten minutes for the preparation test (the banner in the app asks for ten; on an outing with boxes brought from the same room, five are enough). Then read the **scatter**: the largest value minus the smallest. The ΔT card gives it directly, since that is exactly what it computes, and the **Record the scatter** button keeps it for the verdict. A scatter of 0.0–0.4 °C is normal.

What to do with it, at Explorer. One thing, and it is enough: *any field difference smaller than the scatter is not reported as a result.* Write "not detectable with our setup". No further statistics are needed at this level.

What to do with it, at Scientist. In addition, compute each box's mean over the co-location interval and its offset from the group's mean. That offset is the box's *correction* and is subtracted from all of its field readings — an operation that shrinks the residual scatter and leaves smaller differences visible. Then discuss two things: that the offset can depend on temperature, so co-location is redone at every outing, not once for the whole project; and that, after correction, the detection limit does not become zero, but drops to the remaining scatter, which must be re-read.

6. Ready-to-use investigations

Every investigation begins with co-location. It is not a formality: it is the measurement that decides what the class will be allowed to claim at the end.

6.1 Investigation 1: co-location (first, always)

The question. How much do our boxes differ from one another when they are in the same air?

The method. All boxes side by side, on the same table, out of direct sun and any draught, at the same height. Five minutes before every outing; ten minutes for the preparation test the day before. Read the ΔT card at the end of the interval, not at the start, and press **Record the scatter** right then.

What I measure. The scatter, that is ΔT under co-location, in °C, to one decimal place. At Scientist, in addition, each box's mean over the interval.

Expected result. 0.0 to 0.4 °C. Above about 1 °C, a box is dropped from the set or its offset is recorded. This number is written large, on the board and on every sheet, and stays there for the whole lesson: it is the group's **detection limit**.

Technical note — right at the step where the class first reads the ΔT card

The verdict is only as good as the scatter you feed it. Under the card, the app writes a sentence that compares ΔT with the scatter of the boxes side by side:

- **after** pressing **Record the scatter**: *No difference you can detect: ΔT is within the scatter of your boxes side by side* — or *ΔT is k times the scatter of your boxes: a real difference between the places*;
- **before** the press, the app assumes about 1.0 °C, as much as two unscreened boxes often differ, and says so: *Within about 1.0 °C … Measure your boxes' scatter first* — or *Larger than two boxes usually differ side by side … confirm it with the side-by-side test*;
- at any ΔT **above 3 °C** it adds the warning *A large difference: check that no box stands in direct sun. An unshaded box can read several degrees too warm.* **In a schoolyard, a difference of that size between shaded places is almost always radiation error**: check the screens before believing it.

Why it matters. The app does not know whether the boxes really stood side by side when you pressed the button. Pressed too early, before all of them had settled, or after one had already been carried out, the button keeps a scatter that is too large, and then every real difference becomes "no difference you can detect". Pressed on a table with one box in the sun, the same. And without a press, the assumed 1 °C is three times the typical scatter of screened boxes (0.3 °C), so the app will be too cautious.

What to do. Treat the verdict row as a claim to check, not a result — exactly like the pressure row in *Weather Station*. Write the class rule on the board on day one and leave it there: **we compare ΔT with our own scatter, written on the sheet**. For a strong group, the best task is to check the app's sum by hand: $k = \Delta T/\text{scatter}$.

Two more things about the card. ΔT is computed on **the averages of the last minute**, so after moving a box wait at least a minute, and the five minutes of settling remain the rule. And it is computed over **every connected box** — including one forgotten on the teacher's desk, indoors (a disconnected box drops out of the sum). A box left inside on a winter day makes ΔT read a constant 15 °C and erases every difference between the outdoor locations. Count the boxes in the list before reading the number.

6.2 Investigation 2: sun and shade

The question. How much cooler is the shade than the sun, on the same surface and at the same moment?

The method. At least two boxes. One in full sun, one in deep shade — ideally under a tree, otherwise on the shaded side of a building — on the *same* surface and at the *same* height, **both under a white screen** — when the app's banner says "always in the shade", the screen is the box's shade, even on a sunlit spot. A single thing changes: the sun. Not held in hand. Both left for at least five minutes before the first reading. Then read ΔT once a minute, for five minutes.

What I measure. Each box's temperature and their difference. Note the surface, the height, the time and the cloud cover.

Expected result. With **both boxes screened**, on a sunny day, 0.5 to 2 °C cooler in the shade; on an overcast day, often under 0.5 °C. **Without a screen**, the box in the sun reads 3 to 8 °C more — but that is **radiation error**, the casing warmed by the sun, not the air. If ΔT goes above three degrees — the app flags it —, check the screens first. The contrast between the two kinds of weather is exactly the conclusion students should reach: *the shade effect exists only when there is direct sun to block.* On an overcast day, compare the difference with the scatter before saying anything.

6.3 Investigation 3: ranking the surfaces

The question. Which surfaces in our schoolyard are the warmest, and can we rank them?

The method. Four to six surfaces: asphalt, concrete paving, mown grass, bare earth, synthetic sports turf, gravel, a hedge if there is one. Best is *one box on each, simultaneously* — then the weather has no chance to change between readings. If you do not have enough boxes, carry a single one along the route, five minutes at each location, and leave at least one box fixed in the shade as a witness. Height, shading and time budget stay identical throughout, and the time is noted at every stop.

What I measure. Temperature and relative humidity at each location, the time, the surface and the sun exposure.

Expected result. A consistent ranking, with closed, impermeable surfaces the warmest and vegetated, damp ones the coolest, and humidity running the other way — the coolest locations are usually the most humid, which is direct evidence of the evapotranspiration mechanism. In air, with the boxes screened, on a sunny afternoon, differences of 1 to 3 °C.

6.4 Investigation 4: the daily cycle

The question. At what time of day is the difference between our locations largest?

The method. The same route as investigation 3, walked three times on the same day: as early as possible, around noon, and at the end of the school day. If the timetable allows, leave two boxes fixed — one above grass, one above asphalt — recording continuously from morning to evening: you get a whole curve instead of three snapshots. Export after each walk, into separate folders, with the date and time in the name.

What I measure. ΔT between the paired locations at each moment, and the full series if you left fixed boxes running.

Expected result. The difference grows over the day and peaks in mid to late afternoon, not at solar noon — a delay produced by thermal mass. If you can leave the boxes out into the evening, the asphalt location cools more slowly than the grass one, which is the schoolyard-scale version of the nocturnal heat island reported for real cities.

6.5 Investigation 5: does height above the ground matter?

The question. How quickly does the influence of a hot surface weaken with height?

The method. Above a single sunlit asphalt surface, three boxes at different heights: on the ground, at about 50 cm and at 1.2 m, the protocol height, fixed with the tape measure. **All three shaded identically** with white card, so you compare the air, not the effect of sun falling on the casings. Leave them for at least ten minutes. Repeat above grass.

What I measure. The temperature at each height, above both surfaces.

Expected result. A pronounced vertical gradient above the asphalt — often 1 to 3 °C between the ground and 1.2 m — and a much weaker one above the grass. Students discover experimentally why weather standards prescribe a measuring height, and why their own results would have been meaningless if groups had held their boxes at different heights. It is the best possible justification of the protocol rule.

6.6 Investigation 6: from one day to another, and from one kind of weather to another

The question. Does the pattern we found hold on a different day and in different weather?

The method. Repeat exactly the route from investigation 3 on at least three occasions with contrasting conditions: a clear, sunny day, an overcast day, and a day right after rain or with visible wind. Nothing about the route, the order, the times or the sensor heights changes. From the exported files, with dates in their names, build a location × date table.

What I measure. The temperature at each location, on each date, and the route's ΔT on each date. Plus that day's co-location scatter — it is re-measured every time.

Expected result. The *ranking* of locations is remarkably stable — asphalt is almost always the warmest — but the *size* of the differences varies enormously, from several degrees in calm, clear sun to almost nothing under thick cloud or wind. Wind mixes the air and destroys microclimate differences; clouds remove the radiation that produces them. This is the central lesson of the whole project: **a real effect can be present and yet undetectable in the wrong conditions**.

The investigation, in one table

The question. How much warmer is the air above the warmest surface in our schoolyard than above the coolest, at the same moment?
The hypothesis. Written on the sheet **before** any measurement, in two forms, both given: the warmest and coolest locations, with **a number with a unit** for the difference between them; and a second number, *how much we think two boxes placed side by side will differ*. Neither is changed afterwards. The second prediction is the surprising one: almost every class says zero.
What I change. One thing only between two compared boxes: **the underlying surface** *or* **the sun exposure**, not both at once — which is why the locations are chosen in pairs.
What I measure. The air temperature, in °C, to one decimal place, at each box, and the difference between the extremes, as given by the ΔT card. Relative humidity is noted too: it is the witness of evapotranspiration.
What I keep the same. The same height above the ground, measured with the tape, at all boxes; the same white shading at all boxes or at none; the same five minutes of settling before reading; the same boxes from start to finish, with their labels; and the two-metre exclusion circle, because a group of students is itself a source of heat.
How many repeats. Five minutes of co-location at the start of every outing, with all boxes on the same table — this is the repeat that matters, and it is redone at every outing, not once for the whole project. In the field, read ΔT once a minute for five minutes, and take the median value or record the range; a single instantaneous reading is too sensitive to a one-second spike.
What makes a result valid — and what makes it an artefact. Compare **the co-location scatter** with **the difference measured between locations**. **If the difference between locations is smaller than the scatter of boxes placed side by side, the correct conclusion is that the difference is not detectable with your setup — not that it does not exist.** These are two different statements, and the distinction between them is exactly what is being taught here. The remedy is not to look for better numbers in the table, but to increase the signal (a different hour, a clear and calm day, locations that contrast more) or to reduce the noise (longer co-location, individual corrections subtracted, a poor box dropped from the set). **The verdict written by the app does not replace this judgement:** it uses the scatter kept by the button — or, if you never pressed it, an assumed 1 °C —, and it does not know whether the boxes really stood side by side at that moment.

7. Analysis after the activity

7.1 What the data should look like

Comparison	Typical ΔT	Interpretation
Two boxes side by side, in the same air	0.0–0.4 °C	The instruments' scatter: **your detection limit**. Measured at every outing.
Sun versus shade, sunny day, both boxes screened	0.5–2 °C	The real difference in the air.
Sun versus shade, the sunlit box **unscreened**	3–8 °C	**Radiation error**: the casing, not the air. Not a heat island.
Sun versus shade, overcast day	0–0.5 °C	Little direct radiation to block. Compare with the scatter before concluding anything.
Asphalt versus grass, afternoon	1–3 °C, in air	Albedo, thermal mass and evapotranspiration together.
Ground level versus 1.2 m, above asphalt	1–3 °C	The surface heats the air from the bottom up. The reason for the height rule.
Indoors versus outdoors, winter	10–20 °C	A useful check that all boxes work — and the trap that ruins ΔT if a box is left forgotten indoors.
Windy day, any pair of locations	often under 1 °C	Air mixing erases the microclimate. A result, not a failure.
Humidity, grass versus asphalt	5–15 percentage points higher above grass	The witness of evapotranspiration.

Any measured difference smaller than the co-location scatter is not a result. Say it plainly, every time; it is one of the most valuable things students take away from this.

7.2 From ΔT to a conclusion, step by step

1. Record the co-location readings, one for each box, at the end of the five minutes. Worked example, with five boxes: 21.3; 21.4; 21.5; 21.6; 21.3 °C.
2. Compute the **scatter**: the largest minus the smallest. $21.6 - 21.3 = \mathbf{0.3\,°C}$. Write the number on the board and check that it matches the one kept by the **Record the scatter** button. It is the day's detection limit.
3. Record the field readings, all taken at the same moment. Worked example: asphalt in sun 27.9 °C, grass in shade 25.1 °C.
4. Compute the difference: $27.9 - 25.1 = \mathbf{2.8\,°\,C}$.
5. **Compare it with the scatter.** 2.8 against 0.3: the effect is over nine times bigger than the detection limit, so it is a result. Write: *"sunny asphalt was 2.8 °C warmer than shaded grass; our boxes' scatter was 0.3 °C, so the difference is real"*.
6. Do the same for a pair where the effect is small. Worked example, overcast day, grass in sun 18.7 and grass in shade 18.5 °C: the difference is 0.2 °C, so *smaller* than the scatter of 0.3 °C. The correct wording is: *"not detectable with our setup"* — **not** "there is no shade effect" and **not** "it came out zero".
7. **Extension, at Scientist:** compute each box's mean over the co-location interval and its offset from the group's mean, then subtract the offset from all of that box's field readings. Re-read the residual scatter: the detection limit has fallen, but not to zero. Repeat co-location

at the next outing and check whether the offsets stayed the same — if not, they depend on temperature, which is itself a result about the instruments.

7.3 Questions to ask after the activity

1. Two boxes placed on the same bench read 21.3 and 21.6 °C. What does that tell you about the smallest difference you have the right to claim as real from the field data?
2. The asphalt location was 3 °C warmer than the grass one. Name three different physical mechanisms that could each contribute to this, and say how you would try to tell them apart.
3. For the same difference of 0.8 °C, the app first wrote *"Within about 1.0 °C"*, and after you pressed **Record the scatter** (at 0.3 °C) it wrote *"a real difference between the places"*. The temperatures did not change. What did, and which of the two sentences do you believe?
4. The largest difference appeared at 15 o'clock, not at 12, when the sun was highest. Why?
5. Cities are usually warmest compared with the surrounding countryside *at night*, while your walk found the maximum difference in the afternoon. Are the two statements compatible? Explain.
6. On a windy day you got 0.2 °C between all locations. Write that day's correct conclusion in a single sentence, without using the word "failure".
7. The school has money to plant trees in only one part of the schoolyard. Using your data, where should they go, and what extra measurement would strengthen your argument?

7.4 Ideas for assessment along the way

- **Low stakes:** the student colours a printed schoolyard plan with three colours — cool, medium, warm — and writes a sentence explaining the warmest location. No calculation is required.
- **Standard:** a field report with the labelled map, a location × time table, a chart, one identified anomaly and **a conclusion that cites the co-location scatter as the detection limit**. Without that number, the report is incomplete however good the charts look.
- **Extension:** a full survey across several days and different weather, with individual box corrections applied and justified, a quantitative estimate of the cooling achieved through shading, an evaluation of the sampling strategy, and a written recommendation to the school leadership, with options and an estimated cost.

Common pitfalls and misconceptions

Skipping the co-location test. The central mistake of the activity. Without it, students cannot tell a real 0.3 °C difference between locations from a 0.3 °C difference between instruments. It is done *always* first, and redone at every outing.
Reading the app's verdict as the result. The verdict is only as good as the scatter kept by **Record the scatter**: without a press, the app assumes 1 °C; pressed with the boxes unsettled or already apart, it keeps a wrong number. The correct comparison is with the measured scatter written on the sheet, and "no difference you can detect" never means "no difference exists".
A box forgotten indoors. ΔT is computed over *every* connected box. One left on the teacher's desk in winter makes the card read a constant fifteen degrees and completely hides what is happening outside. Count the boxes in the list before reading the number.
Holding the box in your hand. Body heat is worth several degrees. Boxes sit on a stand, not carried while being read.
Not enough settling time. A box brought from a 22 °C classroom into 8 °C air needs whole minutes to catch up. Five minutes minimum at each location, and the reading is taken

at the end of the interval, not on arrival.

Sunlight falling on the casing. Direct light on a box warms the sensor above the real air temperature, by 3 to 8 °C — more than any real difference in the schoolyard. This is radiation error, and it manufactures "heat islands" that do not exist. Exactly why official stations use a white screen. Shade *every* box the same way, or none — but never mixed, since then you are measuring the difference in treatment.

Different heights at different locations. The vertical gradient above hot surfaces is large. One agreed height, measured with the tape, at all boxes.

Confusing air temperature with surface temperature. Students expect the dramatic figures they have read about asphalt. Say the distinction early, or they will mark a correct result as wrong.

Comparing locations measured an hour apart, on a changing day. If a cloud passes between the third and fourth location, the comparison is broken. Prefer simultaneous measurement with several boxes — it is exactly why this app exists; if you still walk the route with one box, walk it twice, in reverse order, and average.

Exporting over the previous export. Files come out named after the box and the date, so two walks on the same day overwrite each other. Each outing gets its own folder.

Rain and condensation. A wet box reads badly and can be damaged. If it starts to rain, the walk ends.

Safety and classroom management

The activity takes a class outdoors, on foot, with equipment, in exactly the sunny weather that gives the best data. **The school's own procedures for outdoor activities — consent, risk assessment and the adult-to-student ratio — apply in full; nothing here replaces them.**

Traffic and crossing. Car parks, delivery ramps and the school gate are the real danger. Reflective vests for all students, a route fixed in advance, and an adult positioned at every point where a vehicle access is crossed. **No measuring location is ever set on a roadway.** If the route leaves the school grounds, the school's full off-site procedures apply: parental consent, a named responsible teacher, a first-aid kit, a charged phone and a headcount at every stop.

Sun and heat. The best days for this investigation are also the days with the highest risk of heat exhaustion and sunburn. Hats, sunscreen, water bottles, breaks in the shade and a shortened route in a heatwave. Students with asthma or heat sensitivity are identified beforehand.

Weather. Do not measure in rain, storms or strong wind: wet equipment is both a data problem and an equipment problem.

Slips and trips. Grassy banks, wet paving and uneven ground, while someone is looking at a clipboard. The rule is said before going out: **whoever is reading or writing stands still.** USB cables never cross a walkway.

Supervision. Groups stay within an adult's sight. With several boxes spread across the grounds, stagger the groups so you can see the whole route, or make the walk a single movement of the whole class.

Unattended equipment. No box is ever left alone. Boxes left outdoors can be taken or damaged; each has a named student responsible for it, and nothing stays out overnight without the building administrator's agreement. For investigation 4, with fixed boxes recording all day, choose visible locations agreed in advance.

Adaptations for pupils with special educational needs

Two boxes, two locations, one single number. This is the support route and it reaches the same scientific conclusion. You choose the locations — one in sun, one in shade — and the student's task is to say which was warmer, and by how much. **The ΔT card gives the number directly, so no subtraction is required.** The student still does the part that really matters: comparing that number with the scatter written on the board and ticking "detectable" or "not detectable". That is exactly the skill being targeted, and it requires no arithmetic.

The scatter, already measured. For students who get lost in preparation, do the co-location yourself before the lesson and **write the scatter on the sheet, already printed**. Do not remove the idea from the lesson to simplify it: only move the measuring work. The student needs to see the number and use it, not necessarily obtain it.

Three real roles, two of which can be done sitting down. Roles rotate at every stop:

- **the detection-limit keeper** — holds the day's scatter and **has the right to reject any group conclusion that claims a difference smaller than it**; this is the role with the greatest scientific weight in the group, because they decide what the group is allowed to write, and it can be done from a chair;
- **the data keeper** — stays at the laptop, records ΔT once a minute with its time, and checks on the spot *how many* boxes appear in the list, to catch a box forgotten indoors;
- **the screen reader** — says out loud what the card shows and which box is the warm one.

Boxes can be carried and placed by a classmate, at the student's direction: "more into the shade" and "a hand's width higher" are experimental decisions too, and whoever makes them is directing the experiment.

Oral answers instead of writing. The hypothesis, the conclusion and the justification of the detection limit can be given **out loud**, with the teacher or a classmate writing them on the sheet. **Ticking a box and circling a word are a fully accepted answer** in place of a sentence.

Language and cognitive support. Write on the board, at the start, the activity's four words — co-location, scatter, detection limit, ΔT — and use them unchanged for the whole lesson. Repeat the same sentence every time: *smaller than the scatter means "not detectable", not "does not exist".* For students sensitive to crowding, the two-metre exclusion circle is an advantage: give them a fixed waiting spot, in the shade, at every stop.

If things do not go to plan

You have a single box. The app cannot do anything with it: the card will simply say **Connect at least two boxes**. Do instead the **guided exercise** *Urban Heat Islands* (single-box), which is built exactly for one box that walks a route and reaches the same physical conclusion, using the closed loop instead of co-location. It is not a weaker backup: it is the other half of the pair, and it is done first anyway.

Rain or wind — the indoor route. Microclimates exist indoors too, and the boxes can be spread out just as well inside. Four locations, measured simultaneously, with exactly the same protocol: the **sunlit window sill**, the **middle of the room**, the **corridor** and a **room without sun** or a north-facing wall. Typical differences are 1 to 4 °C, so comfortably above the scatter, and the mechanisms are the same — received radiation, thermal mass, air that mixes or does not. Co-location is done the same way, before spreading the boxes out. If the boxes also have a light sensor, record how bright each location is too, from the **Live Data** page: the link between received light and temperature is then visible directly, which turns the indoor route into a full lesson, not a consolation.

You have twenty minutes, not fifty. This is exactly the short route, and it is done

indoors. Keep exactly three moments: the written prediction — including the prediction of how much two boxes side by side will differ —, the five-minute co-location that checks it, and the lines separating when one box reaches the sunlit sill. **Do not drop co-location**: it is the lesson, the rest is the illustration.

You have no projector. A semicircle around the laptop, close enough to read the ΔT card, and a **screen reader** named for each moment, who says out loud what the lines are doing: "they have separated", "they are moving together", "the difference is growing". The shape of a curve carries very well in words, and describing it out loud is exactly the skill being targeted.

A box will not settle. Ten minutes after moving outdoors is normal. If only one box keeps drifting after the others have stopped, check its exposure — direct sun on the casing, held in a hand, resting on a warm surface — before suspecting the sensor. If at co-location a box consistently comes out over a degree from the others, drop it from the set for that day and record this: it is a result about the instruments, not an incident to hide.

WiFi does not reach the far end of the schoolyard. The boxes **record nothing on their own**: all that is kept is what reaches the laptop, and the minutes a box spends out of range are missing from the data (on the chart, the box shows as offline). Bring the laptop closer to the distant spots — an adult can carry it along the route —, use a phone as a hotspot, or move the spot closer. For truly distant locations, choose from the start the guided exercise, with one box that walks the route together with the laptop and one fixed box in shade as a witness.

You cannot go outside at all. Use the "What the data should look like" table in this guide as a printed data set, together with a co-location scatter of your choosing — for example 0.3 °C —, and ask students to decide, row by row, which differences can be reported and which cannot. All the reasoning stays the same; only the students' own measurement is lost.

8. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for air temperature and ground-observation standards; explains the fixed measuring height and the white screen, that is exactly two of the class protocol's rules.
- **US Environmental Protection Agency** — `https://www.epa.gov` — search for urban heat islands; the clearest available explanation of causes and solutions, and the source of evidence for students' written recommendation.
- **NASA Global Climate Change** — `https://climate.nasa.gov` — search for land surface temperature in cities; the satellite images make a striking contrast with students' results, measured in air, and once again clarify the distinction between the two quantities.
- **NOAA** — `https://www.noaa.gov` — search for heat-island mapping campaigns and observation standards; NOAA runs volunteer campaigns using exactly the method this activity simplifies.
- **Copernicus** — `https://www.copernicus.eu` — European data on surface temperature and urban climate, to compare the schoolyard with the city and the region.
- **STEM Learning** — `https://www.stem.org.uk` — search for geography fieldwork and data processing; ready-made materials for structuring a survey report.

9. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Because the activity spans two levels, the descriptors read as follows:

- **Measurement practice** — the student does co-location before every outing without being reminded, keeps the same height and the same shading at all boxes, waits the five minutes and notes the time with the number. At Scientist level, in addition: checks the device list before reading ΔT, and redoes co-location at every outing.
- **Data interpretation** — at Explorer, the student correctly states which location was warmer and by how much, and compares that number with the scatter. At Scientist, they build the location × time matrix, apply each box's individual offset, and describe how the size of the effect changes over the day while the ranking stays the same.
- **Scientific reasoning** — the student distinguishes "the difference is not detectable with our setup" from "the difference does not exist", explains why the app's verdict is only as good as the measured scatter it was fed, and distinguishes air temperature from surface temperature.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

10. The student sheet

The sheet students fill in is a separate document, named `heat_islands_with_several_boxes.pdf`, in the folder `docs/en/student_sheets/`. It has two pages — the prediction, the co-location and the walk table on the first; the comparison with the scatter, the chart and the conclusion on the second — so it prints double-sided, one sheet per student. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** any measurement.

Three things worth knowing before printing it. **The prediction includes a number for the instruments' scatter**, asked for before it is measured: almost every class writes zero, and comparing that prediction with the measured 0.3 °C makes half the lesson. **The co-location table sits above the field table**, with one row per box and a box in which the scatter is written large — the same value the **Record the scatter** button keeps — and it must be in front of the student's eyes when they fill in the rest of the sheet. **The sheet's last row is a decision, not a number**: the student ticks whether their difference is detectable or not with the class's setup, and the sentence frame begins with *"our scatter was…"* precisely so the judgement cannot be written without it. The tables are not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.

If the class has not yet done the guided exercise *Urban Heat Islands* (single-box), do that first: the field protocol this sheet assumes — height, shading, waiting time, exclusion circle, the time noted — is learned there, with a single box, and cannot be improvised on the day the class receives six.